Emergence of evolutionary thought

1. **Greek Theories**

Theory of Special Creation: This is the idea that species were created by a supernatural being and are immutable, not evolving over time.

Autobiogenesis: Autobiogenesis is the idea that life arises spontaneously from non-living matter without any involvement of previous life.

Biogenesis: Biogenesis is the scientific theory that life originates only from other living things. It is opposed to the theory of autogenesis.

Transpermiesis: Transpermiesis is the transfer of genetic material between organisms, including the transfer of genetic material between different species. This can lead to the evolution of new species over time.

- **Redi's Experiment**: Francesco Redi conducted an experiment in 1668 to test the theory of spontaneous generation, which held that life could arise spontaneously from non-living matter. He set up jars containing meat, one with a lid and one open, and observed that maggots only appeared in the open jar, not in the sealed jar. This showed that maggots did not arise spontaneously from the meat but from flies that had laid eggs on it.
- **Spallanzani's Experiment**: Lazzaro Spallanzani conducted an experiment in the late 1700s to further test the theory of spontaneous generation. He boiled broth in flasks, some of which were sealed and some left open, and showed that no microorganisms grew in the sealed flasks, while they did grow in the open flasks. This demonstrated that microorganisms only grew if they were introduced from the outside, and not spontaneously generated.
- **Pasteur's Experiment:** Louis Pasteur conducted a series of experiments in the 1860s to definitively prove the theory of biogenesis, that life originates only from other living things. He showed that microorganisms only grow in sterilized broth if air is allowed in, but not if the broth is sealed and kept free of contamination. This showed that microorganisms could not spontaneously generate, but only grow if they were introduced from the outside.

2. Pre Darwin Theories:
   - Theory of uniformitarianism (j Hunton): All animals generated as it is now.
   - Theory of inheritance o acquired character
3. Theory Since Darwin
   - Theory of natural selection (Darwin)
   - Theory of Mutation (Hugo de vries)
   - Modern Theory of evolution (T Dobzhansky, Fisher) – aka Synthetic theory

Natural Selection

- **Overproduction**
- **Struggle for existence**
- **Natural Selection**

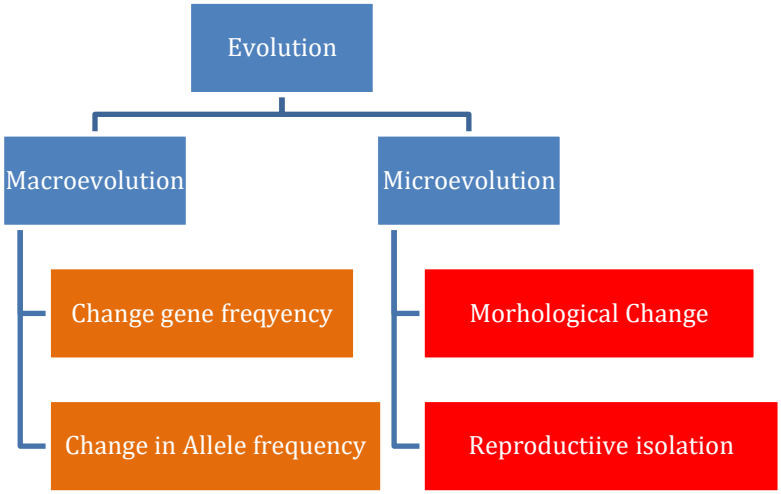
ki

- Tells about survival o the fittest but silent about arrival o the fittest
- Exception is Artificial selection
  - Domestic dog
  - Cultivated crop
  - Ornamental Pigeon
- Sexual Selection
  - Peacock
  - Singing bird
  - Dimorphism

- **Origin of species**
- **Reproductive isolation**
- **Origin of species**

Example:

- Industrial melanism
- Antibiotic plate
- DDT resistant mosquito
- MDR Bacteria



Positive selection: when mutation increase fitness
Negative selection: when mutation decrease fitness or sometimes lethal
Neutral selection: when mutation in have no effect on the organism

$$\frac{dN}{dS} > 1 = Positive\ Selection$$
$$\frac{dN}{dS} > 1 = Negative\ Selection$$
$$\frac{dN}{dS} = 1 = Neutral\ Selection$$

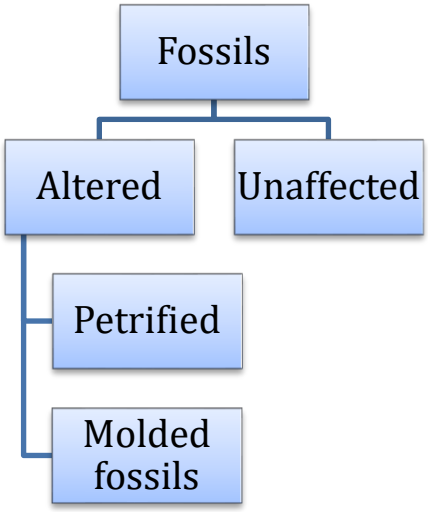
Evidences of evolution

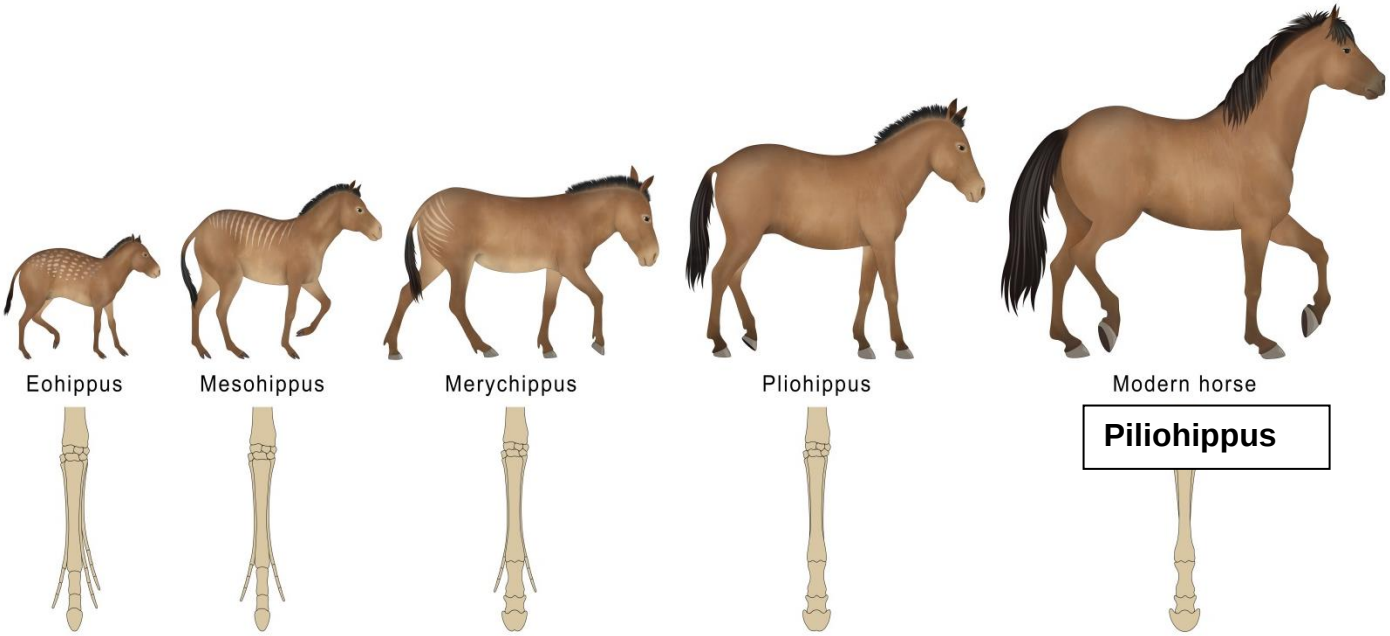
Paleontological Evidence

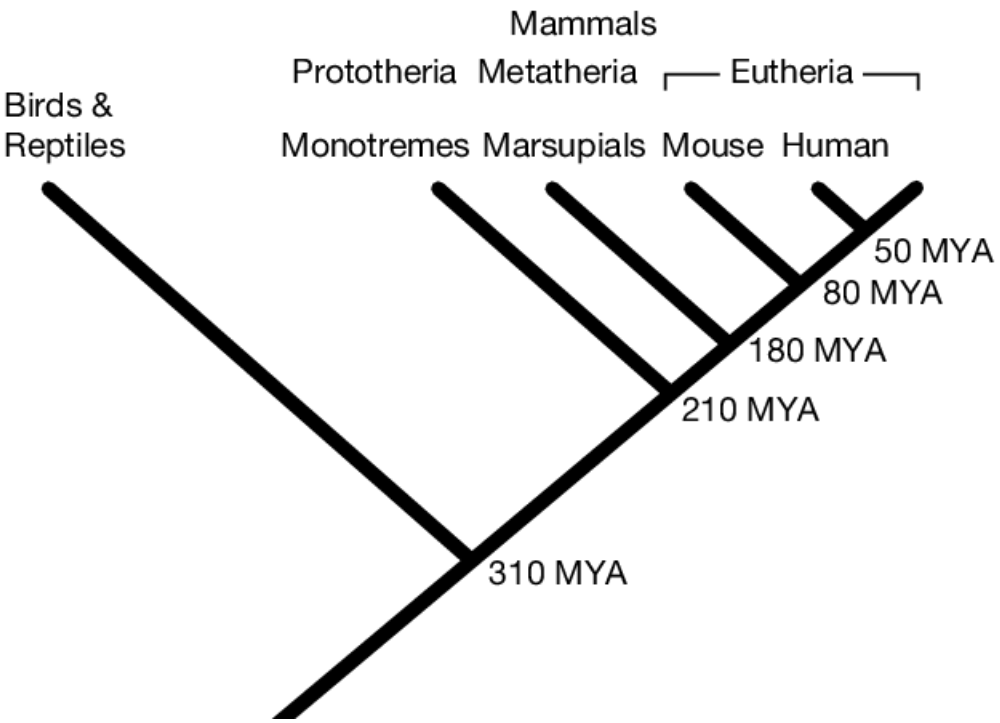
- Fossils
- Sedimentary rock (Connecting link)

Father of palaeontology: **George Cuvier**

ki


Fossils
Altered
Unaffected
Petrified
Molded fossils


Eohippus
Mesohippus
Merychippus
Pliohippus
Modern horse
Piliohippus


Mammals
Prototheria
Metatheria
Eutheria
Birds & Reptiles
Monotremes
Marsupials
Mouse
Human
50 MYA
80 MYA
180 MYA
210 MYA
310 MYA

ki

Comparative anatomy and morphology Evidence

1. Organism
    - Circulatory Evolution
    - Sensory system
    - Digestive system
    - Respiratory system
2. Homologous organ (Divergent Evolution)
3. Analogous character (Convergent Evolution)
4. Vestigial Teeth – Wisdom teeth, Canion in human, Vermiorm Appendix, Hairs on the body
5. Atavism
    - Reversal o ancestral trait
        - Moving ear pinnae
        - Hypertrichosis (Excessive hair growth)

Comparative embryology Evidence

1. Similarity gametogenesis and fertilization
2. Resemblance in young vertebrate embryo (Von Bear) {*Embrodevelopment become increasingly diverse*}
3. **Ontogeny** (Life History) repeats **Phylogeny** (Evolutionary History) (Hackel)
4. Theory of embryonic structure (Larva → Adult)
    - Retrogressive metamorphosis (*Reduction of advanced character during development*)

Bio-geographical Evidence

1. Large combination of continents
    a. Pangia
    b. Lauresia | Gondwana
2. Mammal Evolution and continental drift
    a. **Convergence** of placental (Theria) and marsupial(Metatheria) mammal

Biochemical and molecular Evidence

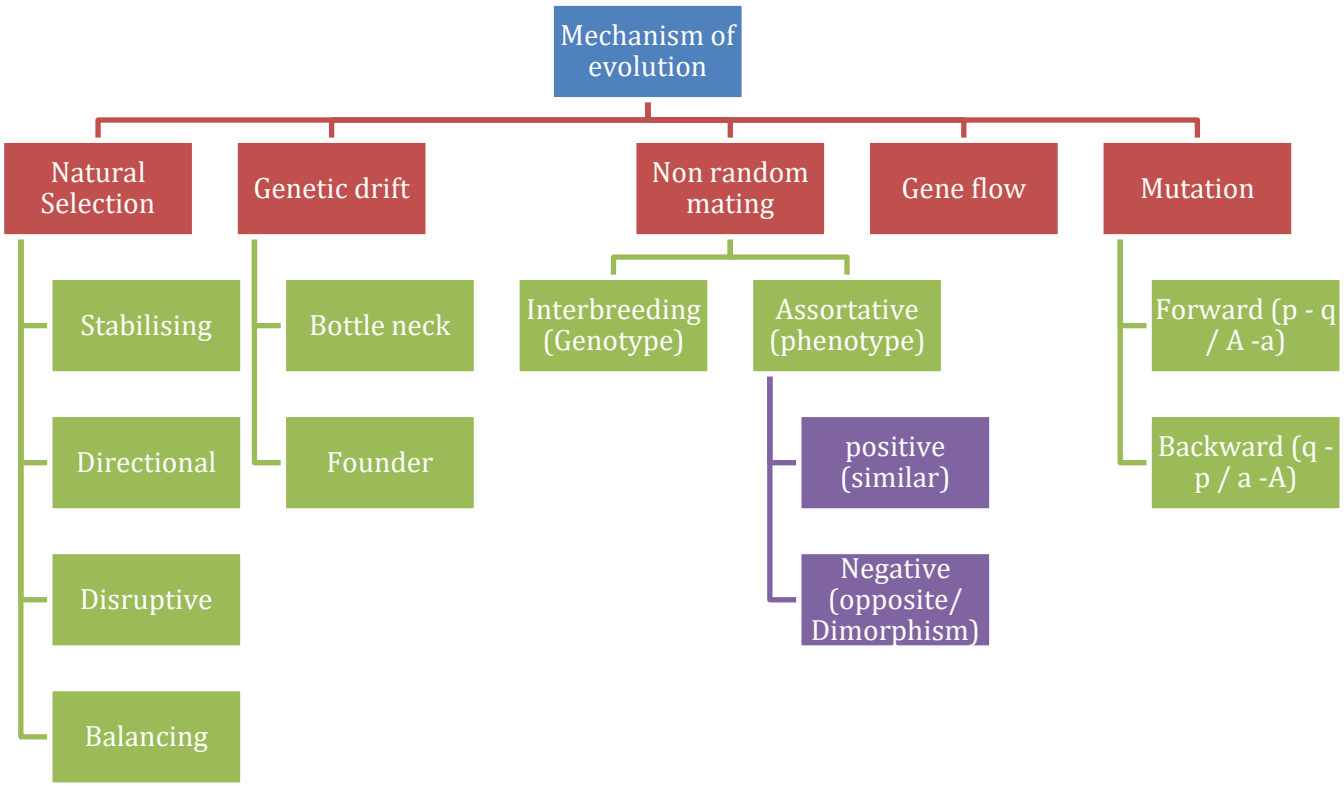
Molecules	Process
1. DNA	Replication
2. RNA	Transcription
3. Amino Acid	Translation
4. Triplet Codon	””
5. ATP	Glycolysis
6. Enzymes	

ki

Gene and allele frequency

Genotype	Number of individual	Gene frequency		Allele frequency
AA	40	P^2	0.4	$A=\frac{2NAA + NAa}{N} = 1$
Aa	20	**2pq**	0.2	
aa	40	q^2	0.4	$A = \frac{2Naa + NAa}{N} = 1$
Total	100			

The mechanism of evolution



- **Natural Selection**

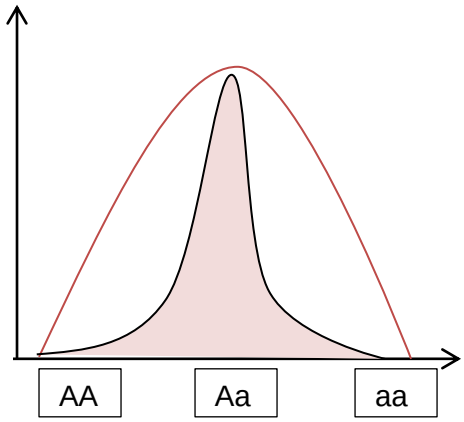
Herdy Weinberg equation

$p^2 + 2pq + q^2 = 1$ (it works only in equilibrium/Non-evolving population)

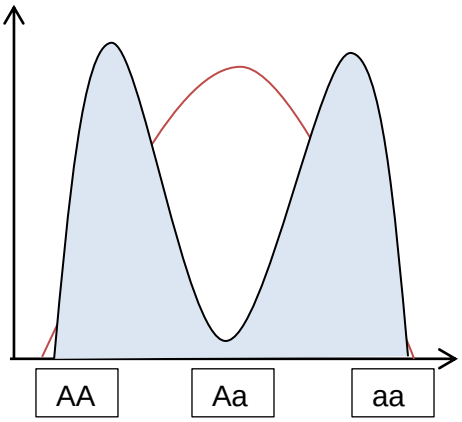
$(p + q)^2 = 1$

Natural selection may biased toward any of the 3 genotype, on the basis of which natural selection can be divided into 4 types

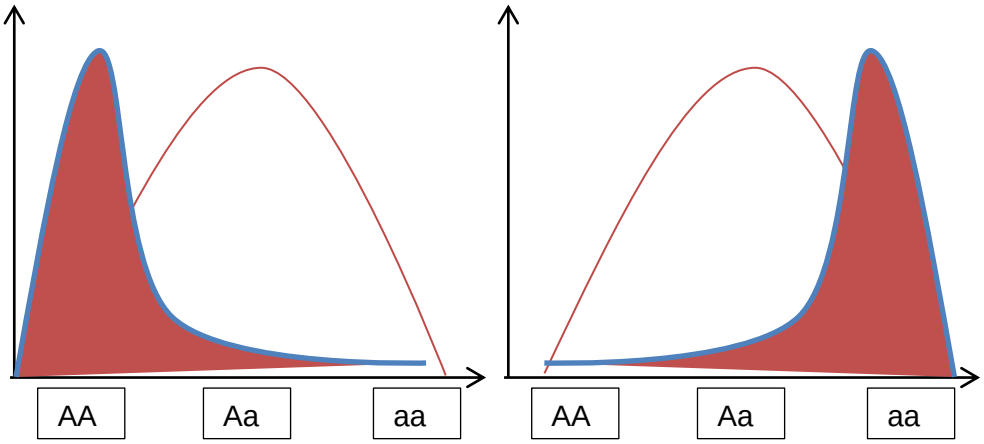
ki



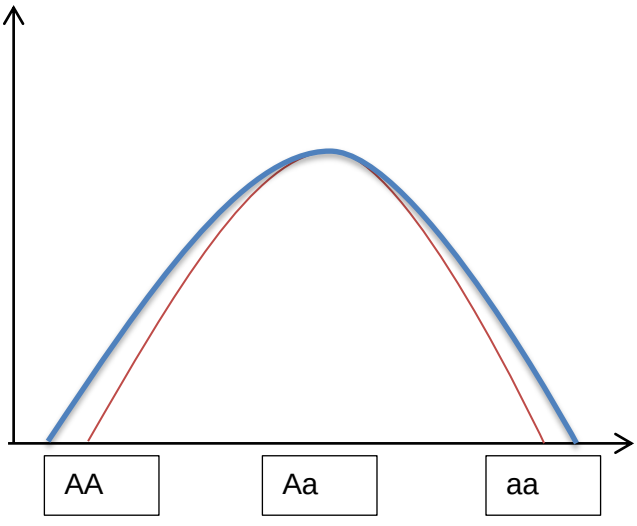
Stabilizing Selection



Disruptive Selection



Directional Selection



Balancing Selection

ki

- **Non random Mating**

Interbreed mating or genotypic mating
Assortative mating or henotypic mating

Inbreeding coefficient (F) = 1 – C

$$= 1 - \frac{Observed\ heterogenity}{Expected\ heterogenity}$$

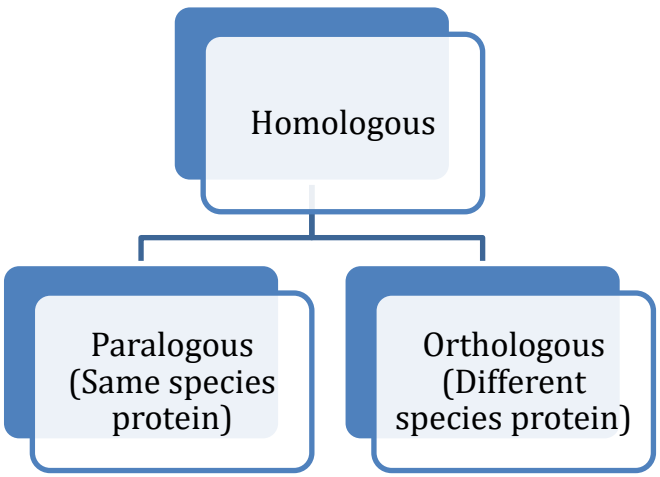
$$= 1 - \frac{Observed\ heterogenity}{2pq}$$

F= 0 means Complete heterogeneous **F = 1** absolute homogenous

▲q = μp – rq
▲p = rq -- μp

$$q = \frac{\mu}{\mu + r}$$

- **Mutation**



4.
5.

ki

Gene creation

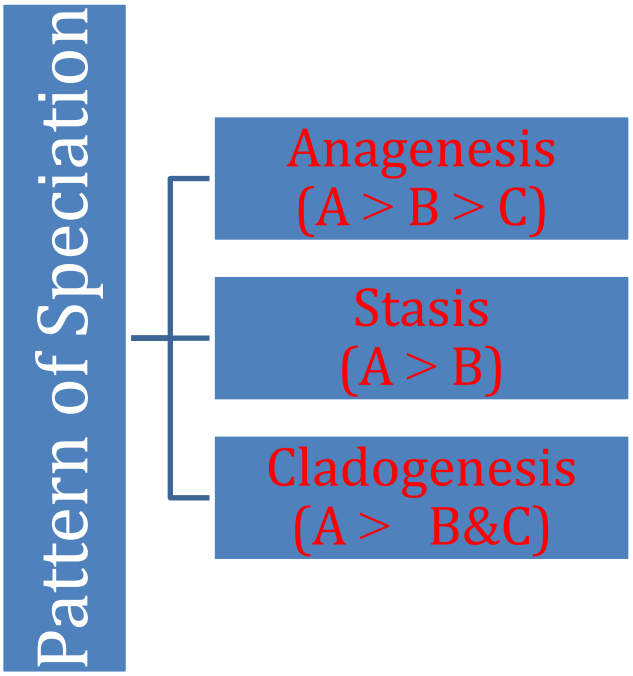
Mutation
Duplication
Retrotransposone
Unequal crossing over
Exon shuffling
Whole genome duplication

Macroevolution

Species Concept

1. Morphological species concept
2. Genetic species concept
3. Phylogenetic species concept
4. Recognition species concept
5. **Biological species concept**

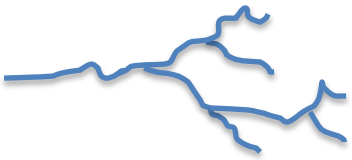
Sibling species: Morpholgically similar species but cannot interbreed

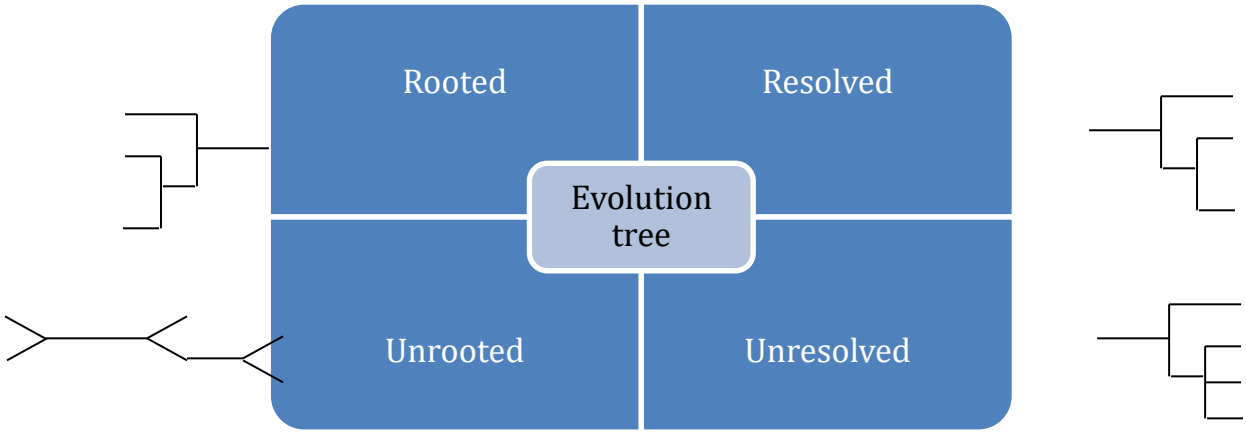


Punctuated Equilibrium	Phyletic gradualism

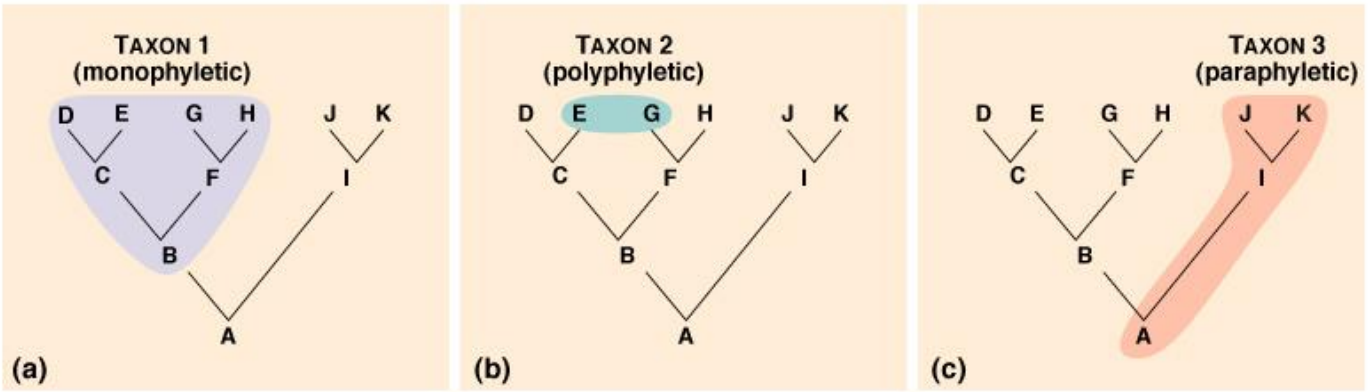
ki

Statis	>>>>>>>>>>>	Stasis





Number of species	No of Unrooted trees	No of rooted trees
3	1	3
4	3	15
5	15	105
6	105	945

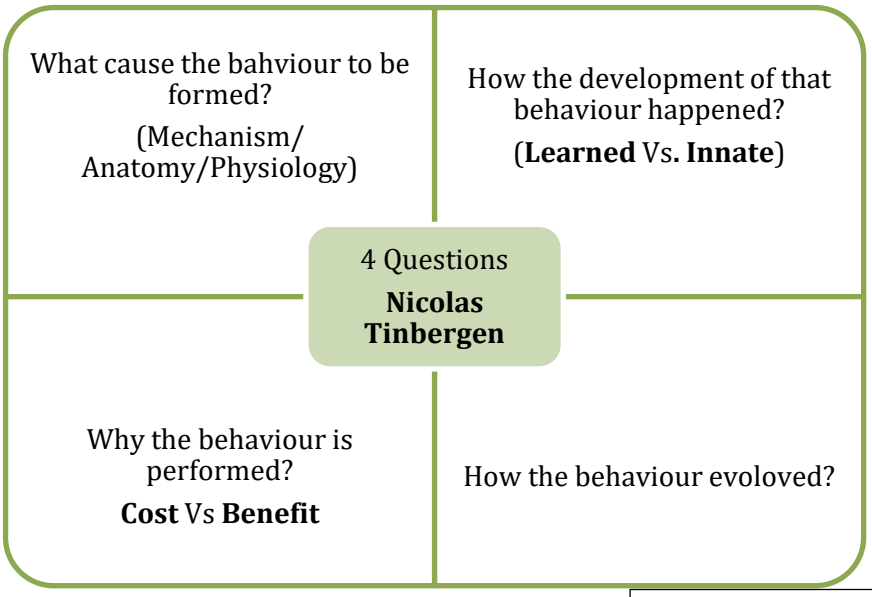


ki

Brain behaviour and evolution

#Causation o Behaviour#

Proximate causation (Mechanism)	Ultimate causation (Mechanism)
Genes Receptors Signalling Organ/Muscle Response	Relative fitness Cost < Benefit



1) Innate Behaviour
   a. No Need of practice
   b. Best performance is in 1st attempt
   c. Stereotype
   d. Genetics involved

- → Kinesis: random movement toward
- → Taxis: Directional movement toward
- → Reflexes: Involuntary response
- → Egg retrieval
- → Egg throwing by cuckoo
- → Defensive response: Fight/Flight

2) Learned behaviour
   a. Need practice, trial and error
   b. No stereotype

- Habituation
- Associated learning
  - Clssical conditioning
  - Operant Conditioning
  - Observational learning
    - Imitation
- Exploratory learning

ki

- **Imprinting**
- **Cognitive Learning**

3) Mixed Behaviour
   a. Innate + Learned
   b. Imprinting: Learning to imitate or follow during critical time period
   c. Spatial learning: Learning to follow space
   d. Cognitive learning: Complex thought and processing involved

EXPERIMENTAL STUDY TO UNDERSTAND ANIMAL BEHAVIOUR

Deprivation experiment

Isolation of young Rabbit just after breeding and grow them on liquid nutrition, after it grows someday it was given a rabbit and it cut and eat it easily without any learning This was a proof of innate behaviour

Hybridization

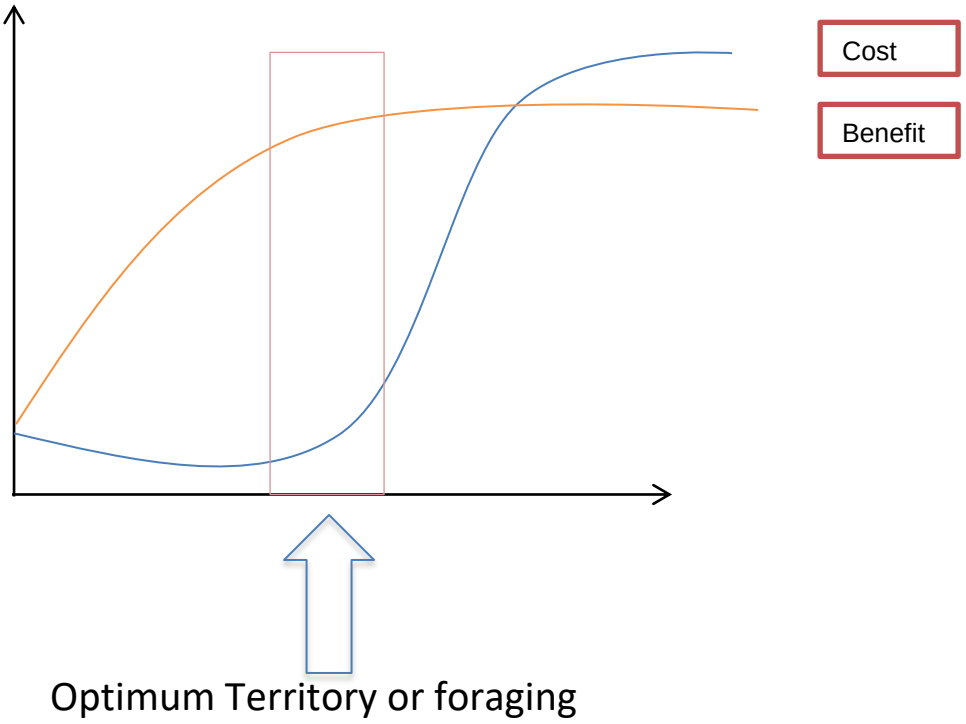
Gene knockout method

Learned s Innate

- Habituation
  Learn to ignore a stimulus
- Associated learning
  Also known as conditioning
  - Classical conditioning
    Association between two different event or stimulus
  - Operant Conditioning / Trial and error
    Association between a behaviour/Output with a stimulus
  - Observational learning
    - Imitation
      Learn to do just by observation no involvement of complex cognition, like baby's facial expression
- Exploratory learning
  Like a chimpanzee use of available tools for a problem solving

ki

Cost Vs. Benefit



Monogamous	Polygyny	Polyandry
Reproductive cost equal	Reproductive more in female	Reproductive more in male
Parental care equal	Parental care more by female	Parental care more by male
Equal selection power and selection pressure	Selection power with female	Selection power is with male
	Selection pressure on male	Selection pressure on female
No sexual Dimorphism	✓ Sexual dimorphism	✓ Sexual dimorphism

Handicap Rule

Reproductive success > Survival Chance

Runaway Hypothesis

Reproductive success = Survival Chance ()

ki